

Square-summable martingale noise without almost-sure convergence

Disproof of conjecture 00000008837

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The final source conjunct states that square summability of martingale-difference noise guarantees almost-sure convergence of stochastic splitting algorithms. Both language versions omit solvability. A genuine backward proximal iteration on an actual probability space disproves that unqualified implication. The separate asserted expected convergence rate is not needed.

Actual operator and resolvent. On the real Hilbert space take

$$A(x) = \{1\} \quad (x \in \mathbb{R}).$$

The graph $G = \{(x, 1) : x \in \mathbb{R}\}$ is monotone since every monotonicity inner product is $(x - y)(1 - 1) = 0$. It is maximal monotone: if a monotone graph containing G contains (u, v) , its inequalities with $(u - 1, 1)$ and $(u + 1, 1)$ give $v - 1 \geq 0$ and $-(v - 1) \geq 0$, hence $v = 1$. Its single-valued representative is globally Lipschitz.

The actual unit-parameter resolvent equation $x \in y + A(y)$ is equivalent to $x = y + 1$. Thus every input has the unique resolvent

$$J_A(x) = (I + A)^{-1}(x) = x - 1.$$

Nevertheless A has no zero. This missing solvability hypothesis is explicitly part of the counterexample.

Probability space, filtration and noise. Let $\Omega = \{*\}$ with its complete measurable space and actual Dirac probability measure $\mu = \delta_*$. For every $n \geq 0$ use the trivial sigma-algebra $\mathcal{F}_n = \{\emptyset, \Omega\}$. This is an actual increasing filtration. Set

$$\xi_n(\omega) = 0 \quad (n \geq 0, \omega \in \Omega).$$

The noise is adapted and integrable, and the actual conditional expectations satisfy

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0$$

for every n . Thus it is martingale-difference noise, with no assumed probabilistic certificate.

Full square summability. At every sample point, and in expectation, the actual squared noise norms vanish:

$$\sum_{n=0}^{\infty} |\xi_n(\omega)|^2 = 0, \quad \sum_{n=0}^{\infty} \mathbb{E}|\xi_n|^2 = 0.$$

Both series are genuinely summable. Consequently the counterexample satisfies either usual pointwise or expected square-summability condition.

Actual stochastic backward iteration. For an arbitrary deterministic initial point x , use the standard additive-noise proximal iteration

$$X_0(\omega) = x, \quad X_{n+1}(\omega) = J_A(X_n(\omega)) + \xi_n(\omega).$$

Every trajectory is adapted. Induction gives

$$X_n(\omega) = x - n \quad \text{for every } n \geq 0, \omega \in \Omega.$$

With zero noise it is also an exact backward inclusion step: $X_n - X_{n+1} = 1 \in A(X_{n+1})$. The unit proximal parameter is strictly positive; all subproblems are defined and uniquely solved.

The actual expectation is likewise $\mathbb{E}X_n = x - n$. Allowing degenerate noise is legitimate here: the universal square-summable-noise assertion does not require nonzero variance or nontrivial sample space.

No sample trajectory converges. If $X_n(\omega) \rightarrow a$ for a finite real a , the shifted sequence has the same limit. Therefore $X_{n+1}(\omega) - X_n(\omega) \rightarrow 0$. This difference is constantly -1 , contradicting uniqueness of limits.

Define the genuine convergence event

$$E_x = \{\omega : \exists a \in \mathbb{R}, X_n(\omega) \rightarrow a\}.$$

The preceding argument gives $E_x = \emptyset$, hence $\mu(E_x) = 0$. Its complement has probability one. In particular almost-sure convergence is false for every start.

Scope and formal correspondence. The disproof concerns the source's unconditional stochastic splitting assertion. It does not contradict a stochastic proximal-point theorem that assumes a nonempty zero set. Lean constructs the actual measure and Mathlib filtration; checks adaptedness, integrability and actual conditional expectations; proves both squared-noise series and actual expectations; verifies the genuine resolvent and full recurrence; and proves the convergence event empty and the almost-sure statement false.

Run `lake build` in `lean/` with Lean 4.19.0 and public Mathlib revision:

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

The project prints the relevant axiom audits.